Research Proposal

On the topic

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A Novel Deep Learning Based Drug-Response Prediction with Multi-Omics, Drug molecules and Gene Interactions

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INDEX

SR. NO.	NAME OF CHAPTER	PAGE NO.
1	ABSTRACT	3
2	INTRODUCTION	3
3	PROBLEM STATEMENT	5
4	STATEMENTS OF OBJECTIVES	6
5	MOTIVATION	6
6	CHALENGES AND LIMITATIONS	6
7	LITERATURE SURVEY	7
8	IDENTIFIED RESEARCH GAP	13
9	PROPOSED RESEARCH METHODOLOGY	14
10	RESEARCH OUTCOME	16
11	PERFORMANCE MEASURES	16
12	CONCLUSION	17
13	REFERENCES	17

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1. ABSTRACT

Drug response prediction is a potential medical research area to improve patients' personalized and targeted treatment strategies by reducing adverse drug reactions. It predicts an individual's response to a specific drug by considering their molecular, genetic, or clinical characteristics. Among the available computational models, deep learning has been identified as a promising drug response prediction method and facilitates multi-omics data integration. This proposal proposed a novel drug response prediction model based on a hybrid deep learning model that comprises pre-processing the multi-omics data, harmonization, imputation, feature selection and data augmentation, and a hybrid deep learning model. The hybrid model considers the graph feature analysis of multi-omics data, transfer learning, and domain knowledge to predict the drug response accurately.

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2. INTRODUCTION

Genetics facilitates a significant role in the complex biological process of human diseases. Even patients with distinct variations and shared similar symptoms impact their response to therapy. Precision medicine considers inter-individual variations in genes, environments, and lifestyles to precisely predict the most suitable treatment for a patient quickly and effectively [1]. Precision medicine has evolved to treat patients with personalized patient care, providing specific treatment based on individual patient characteristics rather than relying on average characteristics over a whole population. In the era of precision medicine, the prediction of personalized drug response can potentially improve the rate of therapy response in various life-endangering disorders [2]. The US Food and Drug Administration (FDA) have been promoted precision medicine approaches to drug development [3].

To advance precision medicine's development, complex biological issues, including drug response prediction, are addressed via several approaches applied to clinical trial data. Drug response prediction is also an essential and challenging issue in bioinformatics and translational medicine [4]. Drug response prediction is key in pharmacogenomics, precision medicine, clinical trials and drug discovery. As a crucial research area, drug response prediction has notable potential for improving patient outcomes with reduced healthcare

ıg response prediction is indispensable in assisting clinicians in making informed regarding the most effective treatments and facilitates the smarter patient selection

and monitoring in clinical trials. Drug response prediction helps diagnose diseases related to psychiatry, cardiovascular disease, infectious diseases, autoimmune disorders, neurological disorders, and oncology. Although, the impact of pharmacogenetics examinations on clinical prescribing practices has been limited, partly owing to the complexity of drug response, which is affected by many factors, including environmental, anthropometric, genetic, and biological subsystems influenced by the disease [5].

With high-throughput molecular and genomic profiling technologies, several drug screening techniques, such as single-cell sequencing and In-silico, have been developed to improve the understanding of disease and treatment [6]. Subsequently, these technologies have rapidly increased the amount of pharmacogenomic data and genomic data generation in genome-wide association research to enhance clinical care via precision medicine further [7]. For potential drug response forecasters, various omic and cell line data such as gene expression and mutation profiles can be assessed. To comprehend large-scale drug screening data sets, computational techniques are essential. While high-throughput screening techniques are used in drug discovery, experimentally screening all potential drugs or drug combinations is impractical and financially infeasible.

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As a consequence, it is requisite to develop computational algorithms for predicting drug response to limit the search space and reduce experimental efforts. More recently, the rapid advancement of computational algorithms has enabled Artificial Intelligence (AI) to build disease models and propel the predictive performance in many tasks of drug discovery and development, such as predicting the effects of treatments, recognizing target genes and functional pathways, and selecting potential biomarkers [8]. Specifically, a subarea of AI, machine learning and deep learning has been explored for drug response prediction to provide individual patient-level predictions and develop a more comprehensive understanding of drug response. Instead of using single-cell sequencing and In-silico techniques in drug response prediction, the advanced computational algorithms can better recognize complex patterns, handle huge-volume of high dimensional, noisy, and missing data and improve performance with reduced cost and time.

Machine learning approaches such as logistic ridge regression, Support Vector Machine (SVM), random forest, Bayesian multitask multiple kernel learning and matrix factorization models have been applied for drug response prediction. Compared to conventional statistical models and machine learning approaches, deep learning performs better in drug response with its high capability of neural networks to capture the complex and non-linear ips among molecular and drug features on the pharmacological and cell-line omics

data [9]. Deep learning-based approaches have been established to learn latent drug features from intrinsic data and impart accurate predictions using its neural architectures, including deep neural networks (DNNs), convolutional neural networks (CNNs), autoencoders (AEs), and recurrent neural networks (RNNs) [10]. Innovatively, deep learning allows researchers to understand cellular drug responses better and discover new therapeutic targets. Hence, the proposed research focuses on utilizing a deep learning-based model for drug response prediction.

3. PROBLEM STATEMENT

In personalized and precision medicine, drug response prediction is important, as it provides treatment decisions and improves patient outcomes. Conventional drug response prediction methods lead to a lack of scalability, an inability to handle high-dimensional data, limited ability to capture non-linear relationships and model interpretability. Deep learning technologies have emerged as a promising solution for drug response prediction by applying deep learning models to forecast how patients will respond to specific drugs. Even so, there is much effort to implement deep learning models that can effectively predict the drug response; to overcome its data complexity and challenges. Variability and high dimensionality are associated with omics data owing to its complexity, making it hard to identify potential variation information and minimize undesirable differences. Unbalanced class distributions lead to over-fitting, particularly for deep neural networks with many parameters and produce inaccurate predictions. For instance, if a dataset has 90% non-responders and 10% responders, a model that solely predicts non-response for each patient would be 90% accurate, although it would be ineffectual for predicting drug response. If the dataset is utilized for training a deep neural network, the network may learn to predict "non-response" perpetually as it will achieve high accuracy; regardless, it does not accurately predict drug response. Integrating multi-omics data considerably enhances the drug response prediction outcome. Still, it faces missing value problems owing to inadequate tissue quality, insufficient sample volume, limitations in measurement systems, or subject dropout. If an important proportion of the data is missing, the drug response model may be unable to learn the underlying patterns and relationships in the data efficiently. In deep learning-based drug response prediction, structural or molecular features of drugs are lost because of fixed molecular representations, leading to suboptimal performance in predicting drug response. netics interaction is due to limited interactions between genes. In other words, the ips between genes and their impacts on drug response are not fully acquired, which

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leads to inaccurate models. Hence, it is necessary to build a novel drug response prediction model using advanced deep learning models with prior domain knowledge to overcome these constraints.

4. STATEMENTS OF OBJECTIVES

The objectives of the proposed research are as follows:

1. To study and compare existing deep learning techniques for drug-response prediction based on multi-omics datasets.
2. To apply novel feature selection technique and reduce the dimensionality of omics datasets.
3. To deal with missing values in multi-omics data using effective imputation method.
4. To experiment optimized dataset with the proposed method of deep learning.
5. To test and validate the performance of the proposed system towards prediction accuracy.

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5. MOTIVATION

Drug response prediction is a potential medical research area to improve patients' personalized and targeted treatment strategies by reducing adverse drug reactions. It predicts an individual's response to a specific drug by considering their molecular, genetic, or clinical characteristics. Among the available computational models, deep learning has been identified as a promising drug response prediction method and facilitates multi-omics data integration.

6. CHALENGES AND LIMITATIONS

Some of the challenges in the proposed model include multi-omics data issues, loss of drug structural features and Sparse Genetic Interactions

- **Multi-Omics Data** in drug response prediction, the datasets suffers from unbalanced class distributions, as it's the number of responders is much lesser than the number of non-responders.
- While incorporating multi-omics data, drug response prediction model confronts missing value, which leads to biasing and imprecise prediction.
- In drug response, deep learning is widely exploited to learn genomic features of cell lines and structural features of drugs.
- But in some cases, deep learning models represent drugs as strings which lead to loss of essential structural information of drug responsiveness.
- If **gene interactions** are ignored, the accuracy of prediction model may be mpromised, and significant details regarding the interaction between genes and ir effect on drug response may be lost.

7. LITERATURE SURVEY

Sr. No.	Paper Titles	Contribution	Research Gap
1	A deep learning model for predicting response to therapy in cancer	Development of deep learning based model to predict drug responses for cancer therapy utilizes gene expression cell line, patient datasets and applies RNA-sequence analysis	A huge volume of supplementary drug response and genomic data will be essential to train inaccurate deep learning models, although wide-ranging valuation must be accomplished on numerous clinical datasets.
2	Predicting drug response of tumors from integrated genomic profiles by deep neural networks	Prediction of the drug response to anti-cancer drugs of tumor by analyzing high dimensional mutations and expressions on pharmacogenomics data using deep learning	High representative authority and model complexity, the method undergoes from over-fitting and the requisite of huge training data. Speak to this, the current study adopts a training–validation partition of Training data to allow early stopping to the training process.
3	MOLI: multi-omics late integration with deep neural networks for drug response prediction	Integration of multi-omics for cancer drug response prediction using deep leaning architecture with transfer learning	Absence of normalization on mutually the input and the output sideways enhances additional encounters to the drug response prediction task.
4	Deep-Resp-Forest: a deep forest model to predict anti-cancer drug response	Prediction of cancer drug response based on ensemble-ensemble algorithm from diverse molecular data	Additional kinds of data will be combined to provide richer evidence for the classification
5	BDKANN-biological domain knowledge-based artificial neural network for drug response prediction	Development of the drug response prediction utilizes the prior biological domain knowledge and Artificial Neural Network (ANN)	Extra data on more drugs and their passageway connection data as it becomes available, growing the predictions for more drugs and cell lines.
6	Prediction of drug response in multilayer networks ı fusion of ics data	Fusion of multi-omics data to predict the response of the cancer cell to therapeutic agents by applying a multilayer similarity network	Deviations between individuals or even within an individual over time expedite variability in omics data, which making it hard to recognize the consistent patterns that are linked with

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			drug response.
7	Predicting drug response and synergy using a deep learning model of human cancer cells	Implementation of modular neural network architecture that merges traditional ANNs with a Visible Neural Network (VNN) to forecast drug response	Incorporate mutations with supplementary levels of molecular evidence such as epigenetic states, gene expression, or micro environmental influences.
8	DeepCDR: a hybrid graph convolutional network for predicting cancer drug response	Combination of multi-omics profiles of cancer cells to predict cancer drug response utilizing hybrid graph convolutional network	High dimensionality in omics data is because of its large scale with numerous data points and this will lead over-fitting problem.
9	SWnet: a deep learning model for drug response prediction from cancer genomic signatures and compound chemical structures	Implementation of novel end-to-end deep learning model to forecast drug response of cancer using multi-task convolutional architecture	In drug response prediction, the datasets suffers from unbalanced class distributions, as it's the number of responders is much lesser than the number of non-responders.
10	Prediction of drug response in major depressive disorder using ensemble of transfer learning with convolutional neural network based on EEG	Design of deep transfer learning strategy to predict drug response from Electroencephalography (EEG) data for Major Depressive Disorder (MDD) by employing an ensemble of CNN-based transfer learning.	Make models open to bias and absence of generalizability.
11	HiDRA hierarchical network for drug response prediction with attention	Prediction of drug response in cancer cell line-drug pairs using hierarchical attention network	Prediction presentation of a model on invisible drugs is not as good as than that on invisible cell lines or invisible pairs.
12	Graph convolutional networks for drug response prediction	Development of drug response prediction by exploiting four variants of Graph Convolutional Network (GCN)	Only genomic data of cell lines was used. Other -omics data such as gene expression not used.
	onvolutional for drug	Analysis for drug response prediction from gene expression	Integration of several types of omics data can improve the

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	response prediction using gene expression data	data using GCN by integrating the Protein-Protein Interaction (PPI) network and gene expression data	knowledge of the status of the cell lines.
14	Deep learning assisted multi-omics integration for survival and drug-response prediction in breast cancer	Implementation of novel deep learning scheme to conduct survival and drug response prediction from the breast cancer data and multi-omics feature integration.	In the some existing investigations, interactions between genes among different omics data types were not contemplated.
15	Graph Transformer for drug response prediction	Prediction of drug response from multi-omics data using graph transformer with Graph Attention Networks (GAT)-GCN	Not discover other techniques to extract an even better drug Representation in Graph Transformer.
16	AGMI: Attention-Guided Multi-omics Integration for Drug Response Prediction with Graph Neural Networks	Integration of multi-omics predict drug response from cell line data by applying attention based Graph Neural Networks (GNNs)	While incorporating multi-omics data, drug response prediction model confronts missing value, which leads to biasing and imprecise prediction.
17	Hybrid quantum neural network for drug response prediction	Development of hybrid quantum neural network to predict drug response of cancer	Depend of quantum neural network.
18	TGSA: protein–protein association-based twin graph neural networks for drug response prediction with similarity augmentation	Drug response prediction by deploying twin GNNs with similarity augmentation to solve scarce data and poor generalization on out of distribution of drugs.	The topology of cell line graphs is assembled basically created on the mutual interaction score of STRING.
19	DualGCN: a dual graph convolutional network model to predict cancer drug response	Implementation of drug response prediction by exploiting drug-GCN module and bio-GCN module with prior knowledge on cancer-correlated genes and protein-protein interactions	If gene interactions are ignored, the accuracy of the prediction model may be compromised, and significant details regarding the interaction between genes and their effect on drug response may be lost.

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20	GADRP: graph convolutional networks and autoencoders for cancer drug response prediction	Implementation of GCN and Auto Encoder (AE) based model to predict cancer drug response	Interactions between genes among different omics data types were not contemplated.

As a state-of-the-art model, the deep learning technology has demonstrated its high capability to predict drug-target interactions, screen for new anti-cancer drugs, and forecast drug synergy in drug discovery, precision medicine and pharmacogenomics research. This section reviews the deep learning approaches for drug response prediction in different applicative tasks.

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7.1 Deep Learning-based Approaches in Drug Response Prediction

A deep learning-based model [11] was developed to predict drug responses for cancer therapy. This model utilizes gene expression cell line and patient datasets and applies RNA-sequence analysis. Subsequently, gene selection and hyper-parameter optimization were performed, and deep neural networks were exploited to predict the drug response and performance validated across multiple clinical cohorts. A novel deep-learning model [12] was proposed to predict the drug response to anti-cancer drugs for tumors. This model analyzes genetic background from high dimensional mutations and expressions using pharmacogenomics data and translates it into predictions of compound treatment response for tumors, using tumor genomic and transcriptomic abstraction techniques. A multi-omics late integration approach [13] was recommended for cancer drug response prediction using deep learning architecture. This approach utilizes somatic mutation, copy number abbreviation and gene expression data to predict the drug response. This approach adopts type-specific encoding sub-networks to learn features for every omics data type, combines learned features into a representation of the multi-omics profiles, and employs transfer learning to increase the training dataset size.

A deep cascade forest model [14] was developed to classify the drug response of cancer based on an ensemble-ensemble algorithm. This model exploits and categorizes the drug response into sensitive and resistive from diverse molecular data. This model transforms raw features, integrates diverse data, and enhances the cascade forest architecture through feature on. Artificial Neural Network based approach [15] was developed to predict the nse of cancer. This approach utilizes the prior biological domain knowledge in the

form of hierarchical connections of genes to drugs to make a deep neural network. This approach predicts multiple drug responses using a deep neural network from the cell line gene expression data. Multi-omics data fusion method [16] was proposed to predict the response of the cancer cell to therapeutic agents. This method constructs a multilayer similarity network to link cell lines and drugs by integrating gene expression profiles, drug chemical structures, copy number variation (CNV), somatic mutation, and drug targets. An interpretable deep learning model [17] was proposed to forecast drug response and synergy of human cancer cells. This model employs a modular neural network architecture that merges traditional ANNs with a Visible Neural Network (VNN) to forecast drug response. The VNN incorporates binary encodings of individual genotypes, and its architecture is affected by a hierarchy of cell subsystems. A hybrid deep learning-based technique [18] was presented to predict cancer drug response by combining multi-omics profiles of cancer cells. This technique adopts a hybrid graph convolutional network comprising a Uniform Graph Convolutional Network (UGCN) to represent the drugs based on the chemical structures of drugs and multiple sub-networks to extract features from multi-omics profiles.

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A novel end-to-end deep learning model [19] was presented to forecast the drug response of cancer using multitask convolutional architecture. By incorporating gene expression, genetic mutation, and cheminformatics, this model applies Graph Neural Network (GNN) to encode the drugs and CNN to extract gene expression and genetic mutation features concurrently. This model also applies a self-attention mechanism to investigate interactions between the genetics and chemical structure of drugs, with the genomic signatures and molecular graphs combined for the efficacy of drug prediction. In the case of Major Depressive Disorder (MDD), an effective deep transfer learning strategy [20] was introduced to predict drug response from Electroencephalography (EEG) data. This strategy uses pre-trained CNNs to predict the outcome of antidepressant treatment for patients with MDD and time-frequency images generated from Wavelet Transform (WT) of pre-treatment resting-state EEG signal. This strategy employs an ensemble of CNN-based transfer learning to classify MDD patients as either responders or non-responders to antidepressants. A hierarchical and interpretable AI-based model [21] was presented to predict the drug response of cancer cell line-drug pairs using drug sensitivity information and gene expression data. This model applies a hierarchical attention network to represent biological pathway and their member genes relevant to drug response. Predicting the drug responses at gene, molecular, evels is beneficial. This model accurately predicts potential drugs that may trigger a response in a specific cancer sample, irrespective of the platform utilized.



A novel drug response prediction method [22] was developed by exploiting Graph Convolutional Network (GCN). This method represents the drugs as molecular graphs to forecast the drug response on cell lines. This method exploits four variants of graph convolutional networks to learn the representative features for each drug-cell line pair, and its fully-connected neural network predicts the drug response value. A new analysis model [23] was introduced for drug response prediction from gene expression data using GCN to learn genomic features. This model generates a gene graph by integrating the Protein-Protein Interaction (PPI) network and gene expression data. Subsequently, GCN identifies the local features, such as sub-networks of genes which are important in the drug response via localized filtering. A novel deep-learning scheme [24] was developed for survival and drug response prediction. This scheme adopts late integrated deep learning to predict breast cancer's survival outcome and drug response from the breast cancer data and multi-omics feature integration. A novel transformer-based approach [25] was proposed to predict drug response from multi-omics data. In this approach, a graph transformer has been developed to enhance the representation of drugs extracted from molecular graphs. This approach predicts drug responses on cell lines by combining graph transformer with Graph Attention Networks (GAT)-GCN.

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A new attention-guided multi-omics integrated technique [26] was proposed to predict drug response from cell line data. This technique builds a multi-edge graph for cell lines and graph edge-aware network that predict drug response by aggregating multi-omics features. This technique applies Graph Neural Networks (GNNs) to integrate multi-omics data and gene constraints for drug response prediction analysis over the whole genome. A novel hybrid quantum neural network-based approach [27] was proposed to predict the drug response of cancer. This approach comprises convolutional, graph convolutional and deep quantum neural layers to predict the response values of drug or cell line pairs. A new drug response prediction model [28] was proposed by deploying twin GNNs with similarity augmentation. This model utilizes GNNs to learn representations of cell lines as graphs based on protein-protein association networks. Similarity augmentation applies GNNs to smooth similar cell lines and drug representations. In addition, an auxiliary pre-training strategy was employed in this model, conducive to solving scarce data and poor generalization on of distribution of drugs.

A dual deep learning-based unified model [29] was introduced for drug response and composed of drug-GCN and bio-GCN modules, respectively. By adopting model predicts cancer drug response by encoding both the chemical structures of

drugs and omics data of cell-line samples. The model integrates prior knowledge of cancer-correlated genes and protein-protein interactions to avoid using large-scale single nucleotide variants data. A GCN and Auto Encoder (AE) based model [30] was introduced to predict cancer drug response. This model utilizes a deep-stacked AE to extract low-dimensional features from cell lines and build a sparse network of drug cell line pairs. By integrating similarity information between drugs, cell lines, and drug cell line pairs, this model adopts an initial residual and layer attention-based GCN (ILGCN) with the fully connected network to make predictions.

8. IDENTIFIED RESEARCH GAP

Even though the deep learning-based strategies outperform the conventional techniques for drug response prediction, some gaps and drawbacks remain unsolved. From the above literature survey, some gaps and drawbacks associated with drug response prediction models are addressed, as given as follows;

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- **Variability, high dimensionality, unbalanced class distributions and missing values:** Multi-omics data are large-scale dataset and highly complex, as it contains several molecular characteristics of patients. Deviations between individuals or even within an individual over time expedite variability in omics data, making it hard to recognize the consistent patterns linked with drug response. High dimensionality in omics data is because of its large scale with numerous data points, leading to over-fitting problems. In drug response prediction, the datasets suffer from unbalanced class distributions, as the number of responders is much less than that of non-responders. This drawback also leads to over-fitting, especially for deep learning models with many parameters. Thus, it is requisite to consider unbalanced class distribution issues in deep learning-based drug response prediction models. Interpretation of multi-omics data in drug response prediction helps to enhance the precision of identifying the effects of drugs on particular cell lines or patient cells. While incorporating multi-omics data, the drug response prediction model confronts missing values, leading to bias and imprecise prediction. Dealing with missing values in multi-omics data will increase efficiency and prediction power.
- **Lose of structural features of drugs:** In drug response, deep learning is widely exploited to learn genomic features of cell lines and structural features of drugs. But some cases, deep learning models represent drugs as strings, leading to the loss of

essential structural information of drug responsiveness. Incorporating molecular graph representations of drugs and multi-omics data for drug response prediction is significant.

- **Sparse Genetic Interactions:** In some existing investigations, interactions between genes among different omics data types were not contemplated. In actuality, genes work in biological networks and interact with one another, leading to potential effects on drug response. If gene interactions are ignored, the accuracy of the prediction model may be compromised, and significant details regarding the interaction between genes and their effect on drug response may be lost.

9. PROPOSED RESEARCH METHODOLOGY

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The proposed research focuses on implementing a robust deep learning-based drug response prediction model. The proposed scheme uses hybrid deep learning techniques to address the multi-omics data-related problems and gene interaction issues. The proposed research model is depicted in below block diagram.

Data Preparation: In data preparation, the multi-omics data is processed to be applied to predict drug response. Data preparation of the proposed work comprises harmonization, imputation, feature selection and data augmentation.

- **Harmonization:** In data preparation of multi-omics data, harmonization is applied as an initial process for making the data in a consistent format and controlling data quality measures. This process helps to minimize the batch effect in multi-omics data.
- **Imputation:** Multi-omics data contains missing values, which can affect the accuracy of the prediction model. Conducting imputation for missing values is necessary during the analysis of multi-omics data. Imputation techniques can be utilized to fill in missing values in the data.
- **Feature Selection:** The feature selection technique will be applied to recognize the most informative features for drug response prediction. In this model, the feature selection technique helps deal with multi-omics data's high dimensionality.
- **Data Augmentation:** The issue of unbalanced class distributions and its effect include, including over-fitting, will be addressed through advanced augmentation techniques. This pre-processing step makes further data analysis more productive by balancing data.

Drug Response Prediction: Subsequently, to predict the drug response, the following processing is incorporated;

- **Graph Feature Analysis:** In this proposed research model, the multi-omics data's structural and genomic features are conducive to improving drug response prediction. For better structural and genomic feature representations, the proposed research model develops graph feature representation at a multi-scale level.
- **Transfer Learning:** The transfer learning concept is integrated into the proposed model to acquire essential gene interaction. Transfer learning helps to adopt the prior biological domain knowledge from gene oncology to capture information about gene interactions.
- **Hybrid Graph-based Deep Learning Model:** The proposed research model develops the hybrid graph-based deep learning model to accurately predict drug response by combing a variety of information, including the domain knowledge and essential graph features of multi-omics data.

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This section provides publicly available datasets, software requirements and performance metrics that will be adopted for the drug response prediction model.

- Some of the benchmark datasets that could be applied for drug response prediction on multi-omics data are Genomics of Drug Sensitivity in Cancer (GDSC) [31], Cancer Cell Line Encyclopedia (CCLE) [32], Cancer Genome Atlas (TCGA) [33] and PubChem [34].
- The software requirements of the experimental model of the proposed drug response prediction model using Python programming language and machine learning libraries.
- The software environment of this experimental analysis is Python executing on Ubuntu 16.04 64-bit operating system machines powered with a 3GHz Intel CPU and 16GB memory to implement deep learning-based drug response prediction.
- Evaluation metrics such as Sensitivity, Specificity, Precision, False Positive Rate, F1 Score and Area under the Precision-Recall Curve will be measured to validate the performance of the proposed research work.



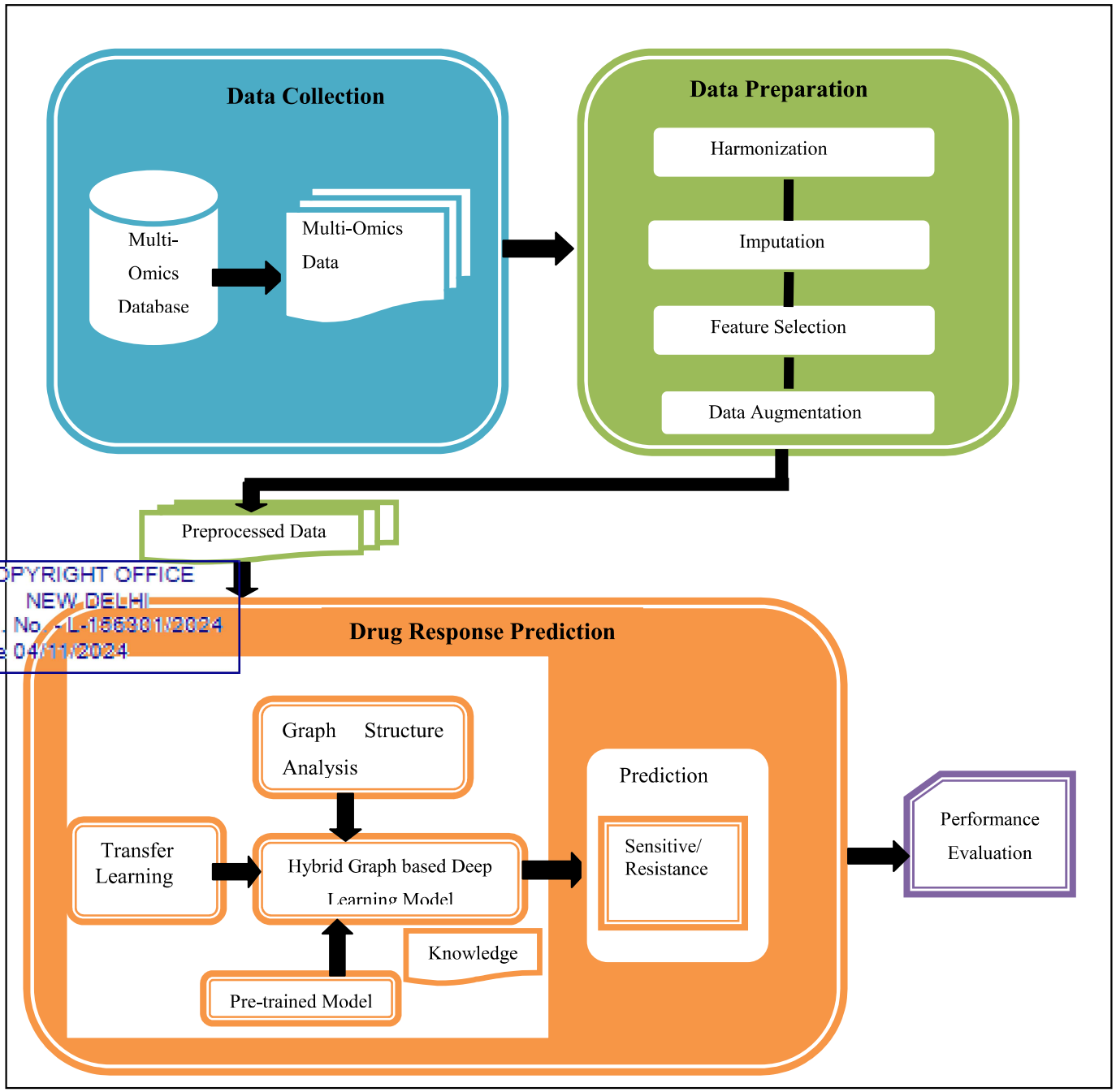


Figure 1.1 Block Diagram of Deep Learning-based Drug Response Prediction

10. RESEARCH OUTCOME

1. Apply the deep learning techniques for drug-response prediction based on multi-omics datasets.
2. Implement novel feature selection technique and reduce the dimensionality of omics datasets.
3. Impute missing values in multi-omics data using effective method.
4. Proposed method for Deep Learning Based Drug-Response Prediction with Multi-
   nics, Drug molecules and Gene Interactions with optimized dataset
   edict the response of drugs through hybrid graph based deep leaning model.

11. PERFORMANCE MEASURES

Evaluation metrics such as Sensitivity, Specificity, Precision, False Positive Rate, F1 Score and Area under the Precision-Recall Curve will be measured to validate the performance of the proposed research work.

12. CONCLUSION

In precision and personalized medicine, drug response prediction is crucial for delivering appropriate treatment and reducing the risk of adverse side effects. Various computational models have been exploited to predict that drug response is beneficial for those involved in advancing precision medicine. Over the past decades, deep learning-based approaches have shown great promise in drug response prediction and enabling multi-omics data integration. However, such a deep learning-based approach faces several challenges in predicting accurate drug responses. Some of the challenges in the proposed model include multi-omics data issues, loss of drug structural features and Sparse Genetic Interactions. A novel model for hybrid deep learning-based drug response prediction was proposed.

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